

Original Research

Determining homologous recombination deficiency scores with whole exome sequencing and their association with responses to neoadjuvant chemotherapy in breast cancer



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ABSTRACT

Recent studies demonstrated that homologous repair deficiency (HRD) score is a useful marker for response to poly (ADP-ribose) polymerase inhibitors or platinum-based chemotherapy. We determined HRD scores and elucidated the clinicopathologic characteristics of HRD-high tumors and their response to non-platinum-based chemotherapy. Primary breast cancer patients ($n = 120$) were pre-operatively treated with paclitaxel followed by 5-fluorouracil/epirubicin/cyclophosphamide (P-FEC). Germline and somatic homologous recombination related gene mutations (gHRRm and sHRRm, respectively) and HRD scores were analyzed using whole exome sequencing (WES) in tumor tissues obtained before chemotherapy. Of 120 tumors, 30 were determined to be HRD-high tumors, significantly associated with high Ki-67 ($P = 0.014$), ER negativity ($P = 0.007$), and PR negativity ($P = 0.021$). Triple-negative cancers showed significantly higher HRD scores than the luminal, luminal-HER2, and HER2 subtypes ($P = 0.023$, 0.016 , and 0.033 , respectively). HRD scores were significantly higher in tumors with gHRRm than in those with sHRRm ($P = 0.002$) or wild-type HRR genes ($P = 1.44 \times 10^{-4}$), but no significant difference was found in HRD scores between tumors with sHRRm and wild-type HRR genes ($P = 0.206$). HRD-high tumors had significantly ($P = 0.003$) higher pCR rates and higher near-pCR rates ($P = 0.049$) compared with those of the HRD-low tumors in all tumors and the luminal subtype, respectively. HRD-high tumors were associated with aggressive phenotypes and gHRRm, but not sHRRm. Our findings suggested that HRD scores might be useful in predicting response to P-FEC in the luminal subtype.

Introduction

Homologous recombination repair (HRR) plays a critical role in the DNA double-stranded break (DSB) repair pathway, and its functional status can be evaluated by homologous repair deficiency (HRD) score which is determined by the sum of the loss of heterozygosity (LOH), telomeric allelic imbalance (TAI), and large scale state transition (LST) scores [1–5]. Almost all breast tumors that arise in BRCA1 or BRCA2 germline mutation carriers are HRD-high tumors. The BRCA1 promoter methylation and TP53 somatic mutations are associated with HRD-high tumors as well [3, 4, 6, 7]. HRD-high tumors are also more prevalent

(40%–60%) in the triple-negative breast cancer (TNBC) subtype than others [4, 8] and are associated with aggressive phenotypes including high histologic grade [4].

Additionally, the association of HRD scores with responses to platinum agents that induce DSB has also been intensively studied in TNBC. The GeparSixto trial demonstrated that no significant difference in pCR rates was found between platinum-based chemotherapy and non-platinum-based chemotherapy in the HRD-low tumors (29.6% vs. 20.0%, respectively) [9]. However, the pCR rates of platinum-based chemotherapy were significantly higher than those of non-platinum-based chemotherapy in the HRD-high tumors were (63.5% vs. 33.9%,

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respectively). Telli et al. reported that HRD-high tumors are associated with a better response to platinum-based chemotherapy in TNBCs [10].

Alternatively, studies on the association of HRD with a response to anthracycline/taxane-based chemotherapy are still limited. Since anthracyclines can induce DSB, HRD-high tumors might be more sensitive to them than HRD-low tumors. Telli et al. showed that HRD-high tumors were more likely to achieve pCR than HRD-low tumors to neoadjuvant chemotherapy (NAC) composed of anthracycline, taxane, or both regimens in TNBC [11], whereas the association of HRD-high tumors with resistance to those regimens was also reported [12]. Sharma et al. evaluated the association of HRD with prognosis of TNBC patients treated with adjuvant doxorubicin + cyclophosphamide and found that HRD-high tumors had a significantly more favorable prognosis than HRD-low tumors [13]. These findings seem to indicate that HRD-high tumors have increased sensitivity of either anthracyclines or taxanes or a combination of both regimens in TNBC.

Although most prevalent in TNBC, 5–10% of luminal subtype tumors are HRD-high tumors [4, 8] and, thus, studying the association of HRD with responses to anthracycline/taxane in the luminal subtype is important. Here, we aimed to clarify the clinical and biological features of HRD-high tumors and determine their association with responses to anthracycline/taxane in the NAC setting, specifically in the luminal subtype.

Materials and methods

Patients and samples

We retrospectively recruited 119 patients with stages II or III breast cancer and a patient with stage IV disease (Table 1). These patients were treated with NAC, a combination of paclitaxel for 12 cycles weekly at 80 mg/m² followed by 5-fluorouracil/epirubicin/cyclophosphamide (FEC, 500/75/500 mg/m²) every three weeks for four cycles at Osaka University Hospital between January 2004 and January 2016. Of the 44 patients with human epidermal growth factor receptor 2 (HER2)-positive tumors, 26 received NAC with trastuzumab (the loading dose of 4 mg/kg followed by 2 mg/kg doses in 12 cycles) concurrently with paclitaxel and 18 received NAC without trastuzumab. Before NAC, all patients underwent vacuum-assisted core-biopsy under ultrasound guidance (Mammotome® 8 G; Devicor Medical Products, Cincinnati, OH). The biopsied tumor samples for histologic examination were fixed in 10% buffered formaldehyde and those for whole exome sequencing (WES) and microarray analysis (OncoScan®) were snap-frozen in liquid nitrogen and kept at −80 °C until use. Peripheral blood was also collected from each patient and peripheral blood leukocytes (PBL) were separated for WES analysis of germline DNA. This study was approved by the Osaka University Research Ethics Committee.

Whole exome sequencing

Libraries for WES were prepared from DNA isolated from tumors and PBL matched using the SureSelect XT Low Input Target Enrichment System® (Agilent Technologies Inc., Santa Clara, CA), the SureSelect Human All Exon V6® (Agilent Technologies Inc., Santa Clara, CA), and the SureSelect XT Low Input Dual Index P5 Indexed Adapters® (Agilent Technologies Inc., Santa Clara, CA) according to the manufacturer's instructions. WES for the 120 breast tumor and matched PBL pairs was carried out on the Illumina HiSeq 3000 platform® (Illumina Inc., San Diego, CA), generating 2×101 read pairs with a 60×(germline) or 120×(tumor) coverage.

Data processing

The quality of sequence reads was assessed with FastQC [14]. Poor-quality bases, low quality reads, and the adapter sequences were removed using Trim Galore! (v0.5.0) with the following command

Table 1
Clinicopathologic characteristics and HRD score.

Characteristic	Total N = 120, (%)	HRD score		P value
		High N = 30, (%)	Low N = 90, (%)	
Age				
> 50 years	65(54.2%)	17 (56.7%)	48 (53.3%)	0.751
≤ 50 years	55 (45.8%)	13 (43.3%)	42 (46.7%)	
cT				
1	14 (11.7%)	5 (16.7%)	9 (10.0%)	0.451
2	81 (67.5%)	21 (70.0%)	60 (66.7%)	
3	13 (10.8%)	1 (3.3%)	12 (13.3%)	
4	12 (10.0%)	3 (10.0%)	9 (10.0%)	
cN				
0	41 (34.2%)	16 (53.3%)	25 (27.8%)	0.010
1	76 (63.3%)	13 (43.3%)	63 (70.0%)	
2	1 (0.8%)	0 (0%)	1 (1.1%)	
3	2 (1.7%)	1 (3.3%)	1 (1.1%)	
HG				
1	20 (16.7%)	1 (3.3%)	19 (21.1%)	0.066
2	55 (45.8%)	15 (50.0%)	40 (44.4%)	
3	45 (37.5%)	14 (46.7%)	31 (34.4%)	
Ki-67				
≥ 20%	79 (67.5%)	26 (86.7%)	53 (60.9%)	0.014
< 20%	38 (32.5%)	4 (13.3%)	34 (39.1%)	
ER				
Positive	84 (70.0%)	15 (50.0%)	69 (76.7%)	0.007
Negative	36 (30.0%)	15 (50.0%)	21 (23.3%)	
PR				
Positive	70 (58.3%)	12 (40.0%)	58 (64.4%)	0.021
Negative	50 (41.7%)	18 (60.0%)	32 (35.6%)	
HER2				
Positive	44 (36.7%)	10 (33.3%)	34 (37.8%)	0.662
Negative	76 (63.3%)	20 (66.7%)	56 (62.2%)	
TILs				
High	43 (35.8%)	14 (46.7%)	29 (32.2%)	0.156
Low	77 (64.2%)	16 (53.3%)	61 (67.8%)	

cT, clinical tumor factor; cN, clinical nodal status; HG, histologic grade; TILs, Tumor-infiltrating lymphocytes; HRD, homologous recombination deficiency.

line: trim_galore -quality 20 -phred33 -stringency 3 -gzip -length 50 -paired -output_dir <read1.fastq.gz> <read2.fastq.gz> [15]. The trimmed sequence reads were aligned to the reference human genome (GRCh38) and the decoy sequences for Japanese reference genome (JRG) version2 (sequences assembly to GRCh38; decoyJRGv2) from JRG (<http://jrg.megabank.tohoku.ac.jp/>) using Burrows-Wheeler Aligner (BWA, v0.7.17) in the mem mode with default parameters [16]. After duplicate marking and removal of PCR duplicates, base quality recalibration was performed with known-sites to dbSNP build 146 and the Mills + 1000 Genome (1000 G) gold standard insertions and deletions (indels) by the Genome Analysis Toolkit (GATK, v4.1.1.0) and Picard (<http://broadinstitute.github.io/picard/>) [17, 18]. Germline variants were identified using the GATK HaplotypeCaller in the GVCF mode, GenotypeGVCFs, and VariantRecalibrator according to the GATK best practices for germline short variant discovery (<https://gatkforums.broadinstitute.org/gatk/discussion/11145/germline-short-variant-discovery-snp-indels>). Somatic single-nucleotide variants (SNVs) and small indels in tumors were detected using MuTect2 [19].

Variant annotation and filtering

The functional annotation of the variants was performed using the ANNOVAR pipeline [20]. In this study, the HRR genes ($n = 102$) were defined as reported by Riaz et al. and curated appropriately to improve their annotations (Suppl. Table 1) [21]. For germline variants, SNVs and indels found in either germline or tumor, with an overall depth of coverage (DP) < 10 reads, a genotype quality (GQ) < 30, a variant allele frequency (VAF) < 20% in the germline, or not classified in ClinVar as pathogenic (P)/likely pathogenic (LP) were filtered out [22].

For somatic variants, SNVs and indels found in both germline and tumor, with an overall DP < 30 reads, GQ < 30, VAF < 5%, or variant reads < 5 were filtered out. Any germline or somatic variants in the non-exonic regions, synonymous single nucleotide variants, and identical to an entry in dbSNP 146 “common” ($\geq 1\%$ frequency) and $\geq 1\%$ frequency across all Exome Aggregation Consortium East Asia database (ExAC-EAS), 1000 G, Genome Aggregation Database (gnomAD), and Human Genetic Variation Database (HGVD) for Japanese genetic variation (<http://www.hgvd.genome.med.kyoto-u.ac.jp/>) were filtered out. The remaining germline and somatic variants of the HRR genes after the filtering process were defined as germline and somatic mutations, respectively.

Cellularity estimation and LOH identification

Evaluation of tumor cellularity and LOH was done by WES using PureCN (<http://bioconductor.org/packages/PureCN/>) implemented by the ABSOLUTE algorithm [23, 24]. BAM files of tumors, normal control coverages, and VCF files containing allelic fractions of germline and somatic variants by MuTect (v1.1.5) were applied to PureCN using a default parameter [25]. Median tumor cellularity was 42% (range of 10% to 99%) for 120 tumors assayed by WES and 56% (range of 20% to 98%) for 47 tumors assayed by both WES and OncoScan®.

HRD score

The HRD score was determined as a simple sum of the three factors (NtAI, LST, and HRD-LOH), and tumors with HRD score ≥ 42 were defined as HRD-high. For the determination of HRD scores using WES, BAM files of tumor samples were applied to the Sequenza followed by the scarHRD R package with a default parameter as previously described (<https://github.com/sztup/scarHRD>) [26, 27]. For the determination of HRD scores using OncoScan®, copy number variation (CNV) was analyzed by ASCAT (allele-specific copy number analysis of tumors) [28] followed by R code described by Marquard et al. [3], as previously described [12].

Hormone receptors, HER2, Ki-67, and TILs status

Immunohistochemical assay for estrogen receptor (ER), progesterone receptor (PR), and Ki-67 was done as previously described with cutoff values of 1% for ER and PR and 20% for Ki-67 [12]. Representative expressions of these markers are shown in Suppl. Fig. 1. HER2 was analyzed by FISH according to the ASCO/CAP 2018 guideline [29] and tumor-infiltrating lymphocytes (TILs) were evaluated using HE sections according to the International TILs Working Group 2014 [30]. Tumors were classified into four subtypes as follows: luminal subtype, ER(+) and/or PR(+) and HER2(-); luminal-HER2 subtype, ER(+) and/or PR(+) and HER2(+); HER2 subtype, ER(-), PR(-) and HER2(+); TNBC, ER(-), PR(-) and HER2(-). pCR for NAC was defined as no invasive foci in the tumor (grade 3) without lymph node involvement, and near-pCR as grade 3 with lymph node involvement or minimal invasive foci in the tumor (grade 2b) regardless of lymph node involvement. Pathological response for breast tumors was assessed according to the criteria of the Japanese Breast Cancer Society [31].

Statistical analysis

All statistical analyses were performed using the software R version 3.6.3. Pearson's correlation coefficient was used to evaluate the intra-device association between paired DNA microarray and WES. The associations between the various parameters were compared using the chi-square and Fisher's exact tests. The Wilcoxon signed-rank test was used to compare the HRD scores in breast cancer subtypes. The logistic regression model was used for the univariate analysis of the association of the responses of various parameters to NAC. All tests were two-sided and $P < 0.05$ was considered as statistically significant.

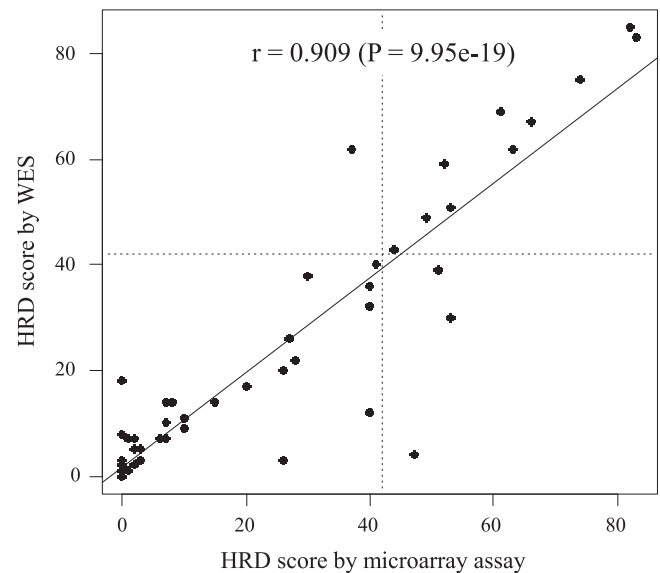


Fig. 1. HRD score correlation determined using whole exome sequencing and microarray assay (OncoScan®). HRD scores determined using whole exome sequencing and OncoScan® in 47 tumors with $\geq 20\%$ cellularity showed a strong correlation (Pearson $r = 0.909$, $P = 9.95e-19$).

Results

Correlation of HRD scores determined by OncoScan® and WES

First, we validated the use of WES instead of the original use of an SNP (single nucleotide polymorphism) array in the determination of HRD scores. Forty-seven tumors were analyzed by both WES and OncoScan® for the determination of HRD scores. Their correlation was excellent with a Pearson $r = 0.909$ ($P = 9.95e-19$; Fig. 1). A high concordance rate (91.5%, 43/47) of HRD-high and HRD-low tumors was also obtained between the two assays. These results confirmed that WES accurately determined the HRD scores as previously reported by Sztupin-szki et al. [27].

Clinicopathologic characteristics of HRD-high tumors

Of 120 tumors, 30 were determined to be HRD-high status. The clinicopathologic characteristics of HRD-high tumors are shown in Table 1. HRD-high tumors were significantly more likely to be clinical N0 ($P = 0.010$), Ki-67 high ($P = 0.014$), ER negative ($P = 0.007$), and PR negative ($P = 0.021$) tumors than HRD-low tumors. Subtype analysis revealed that TNBC had significantly higher HRD scores than the luminal ($P = 0.023$), luminal-HER2 ($P = 0.016$), and HER2 subtypes ($P = 0.033$) (Fig. 2). The incidence of HRD-high tumors was also significantly higher in TNBC (66.7% (10/15)) than in the luminal (16.4% (10/61), $P = 2.78e-4$), luminal-HER2 (21.7% (5/23), $P = 0.015$), and HER2 subtypes (23.8% (5/21), $P = 0.026$). HRD scores were not significantly different among the luminal, luminal-HER2, and HER2 subtypes.

Incidence of germline or somatic mutation of HRR genes

A list of the HRR genes for mutation analysis is shown in Suppl. Table 1. Twelve patients harbored 13 germline mutations of HRR genes including BRCA2 ($n = 8$), BRCA1 ($n = 2$), RAD54B ($n = 2$), and PALB2 ($n = 1$) (Table 2 and Suppl. Table 2). One patient had both BRCA2 and RAD54B germline mutations. Following the exclusion of these 12 patients, somatic mutations of HRR genes were analyzed which revealed that 27 patients had 40 somatic mutations in the 29 HRR genes as shown in Table 2 and Suppl. Table 3.

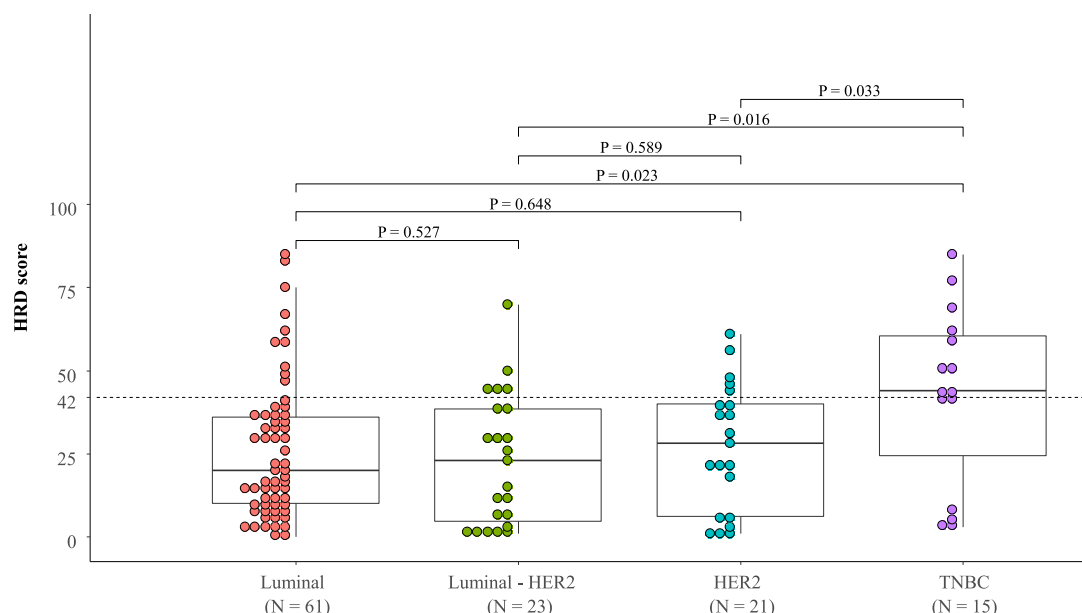


Fig. 2. HRD scores according to breast cancer subtypes. HRD scores determined using whole exome sequencing were compared according to the breast cancer subtypes, i.e., Luminal, Luminal-HER2, HER2, and TNBC (triple-negative breast cancer).

Table 2

HRR genes with germline- or somatic-variants and their frequencies.

Germline variant*			Somatic variant**		
Gene	No. of tumors	%	Gene	No. of tumors	%
BRCA1	2	1.7	ATM	3	2.8
BRCA2	8	6.7	ATR	2	1.9
PALB2	1	0.8	ATRIP	1	0.9
RAD54B	2	1.7	BRCA1	1	0.9
			BRIP1	1	0.9
			CDK12	1	0.9
			CHEK2	1	0.9
			ERCC4	1	0.9
			EXO1	1	0.9
			FAAP100	1	0.9
			FANCB	1	0.9
			FANCG	2	1.9
			FANCM	1	0.9
			MUS81	2	1.9
			NONO	1	0.9
			PALB2	1	0.9
			POLQ	2	1.9
			RAD50	1	0.9
			RAD51B	1	0.9
			RECQL4	5	4.6
			RIF1	1	0.9
			SETMAR	1	0.9
			SFPQ	1	0.9
			SLX4	1	0.9
			SMC1A	1	0.9
			TOPBP1	2	1.9
			TP53BP1	1	0.9
			USP11	1	0.9
			WRN	1	0.9

* The germline variants in HRR genes were analyzed in all tumors ($N=120$). One tumor had germline variants in both BRCA2 and RAD54B.

** The somatic variants in HRR genes were analyzed in tumors without germline-HRR gene variants ($N=108$).

Germline or somatic mutation of HRR genes and HRD scores

HRD scores were significantly higher in the germline HRR gene mutation (gHRRm) group than the somatic HRR gene mutation (sHRRm)

Table 3

Univariate analysis of multiple clinicopathologic parameters in the relation to near-pCR to NAC in the luminal subtype.

Characteristic	N	Grade		Univariate		
		near-pCR	%	OR	95% CI	P value
Age						
≤ 50 years	34	6	(17.6%)	1		
> 50 years	27	4	(14.8%)	0.81	0.19 - 3.19	0.767
cT						
≥ 2	54	9	(16.7%)	1		
1	7	1	(14.3%)	0.83	0.04 - 5.76	0.873
cN						
Positive	44	6	(13.6%)	1		
Negative	17	4	(23.5%)	1.95	0.44 - 7.96	0.355
HG						
1 or 2	42	5	(11.9%)	1		
3	19	5	(26.3%)	2.64	0.65 - 10.92	0.169
Ki-67						
< 20%	26	2	(7.7%)	1		
≥ 20%	34	8	(23.5%)	3.69	0.82 - 26.12	0.120
PR						
Positive	53	8	(15.1%)	1		
Negative	8	2	(25.0%)	1.88	0.25 - 9.99	0.486
TILs						
Low	46	6	(13.0%)	1		
High	15	4	(26.7%)	2.42	0.54 - 10.11	0.225
HRD						
Low	51	6	(11.8%)	1		
High	10	4	(40.0%)	5.00	1.04 - 23.50	0.039

NAC, neoadjuvant chemotherapy; cT, clinical tumor size; cN, clinical nodal status; ER, estrogen receptor; PR, progesterone receptor; HER2, human epidermal growth factor receptor 2; HG, histologic grade; TILs, tumor-infiltrating lymphocytes; HRD, homologous recombination deficiency; OR, odds ratio; CI, confidence interval.

group ($P=0.002$) and the wild-type HRR gene (HRRw) group ($P=1.44e-4$) (Fig. 3). The incidence of HRD-high tumors was also significantly higher in the gHRRm group (66.7% (8/12)) than in the sHRRm (22.2% (6/27), $P=0.012$) and the HRRw groups (19.8% (16/81), $P=0.002$). No significant difference was found in the HRD scores or the incidence of HRD-high tumors between the sHRRm and the HRRw groups (Fig. 3).

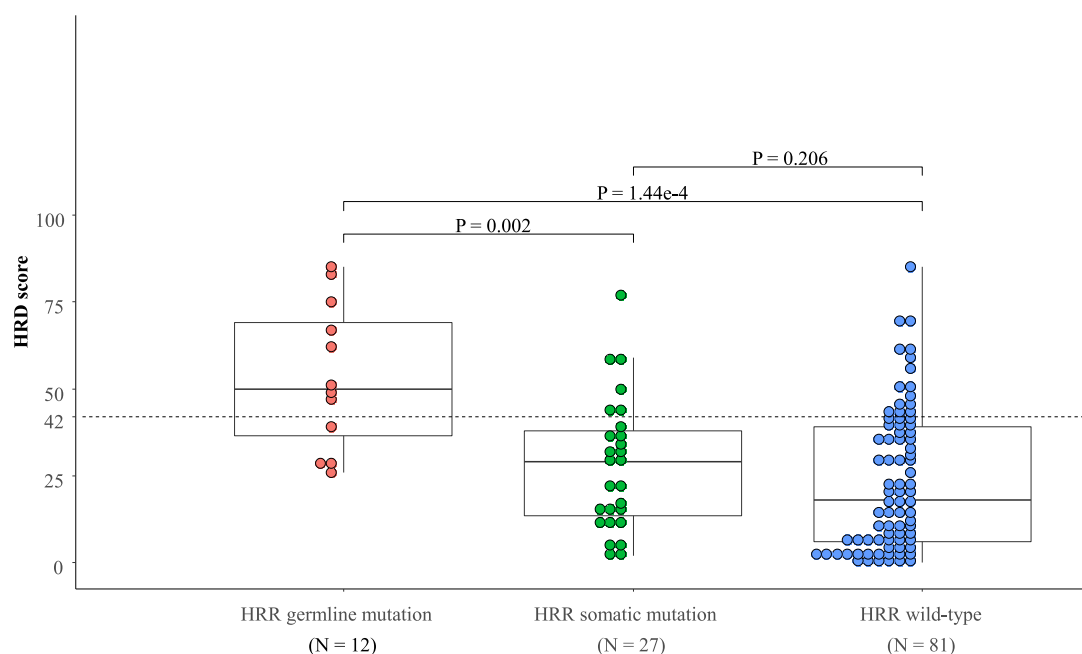


Fig. 3. HRD scores according to HRR gene mutations. HRD scores were compared across tumors with germline ($n = 12$) and somatic ($n = 27$) HRR gene mutations, and wild-type HRR genes ($n = 81$).

Association of HRD score with responses to NAC

Association of HRD score with responses to NAC was evaluated in the 94 patients treated with chemotherapy alone. Patients with HRD-high tumors were significantly ($P = 0.003$) more likely to obtain pCR (39.1% (9/23)) than those with HRD-low tumors (9.9% (7/71); Fig. 4A). The subtype analysis revealed that HRD-high tumors tended to be associated with a numerically higher pCR rate compared with that of the HRD-low tumors (pCR rates: 20.0% vs. 3.9%, $P = 0.122$) in the luminal subtype. However, such a tendency was not observed in the HER2 subtype (luminal HER2 plus HER2 subtype) and TNBC (Fig. 4B). We further analyzed the association between near-pCR and HRD scores. Patients with HRD-high tumors were significantly ($P = 7.32 \times 10^{-4}$) more likely to obtain near-pCR (65.2% (15/23)) than those with HRD-low tumors (23.9% (17/71); Fig. 4C). The subtype analysis revealed that HRD-high tumors were significantly ($P = 0.049$) associated with a higher near-pCR rate compared with that of the HRD-low tumors (response rates; 40.0% vs. 11.8%) only in the luminal subtype (Fig. 4D).

Twenty-six patients with HER2-positive tumors, including a patient with stage IV disease, were treated with NAC and trastuzumab. pCR rates were similar between the HRD-high and -low tumors (pCR rate: 71.4% (5/7) vs. 73.7% (14/19)).

Discussion

HRD scores were originally measured by SNP arrays, whereas the recent study by Sztupinski et al. reported a good correlation between WES and SNP array in the determination of HRD scores [27]. Here, we first attempted to confirm their observation and were able to show an excellent correlation between HRD scores assessed using WES and OncoScan®. Then, we studied the clinicopathologic characteristics of HRD-high tumors detected by WES. Consistently with the previous reports [4, 8], HRD-high tumors were associated with biologically aggressive phenotypes such as high Ki-67 and negative ER/PR and were most prevalent in TNBCs. A significant association of high HRD scores with cN0 seems to be attributable to a higher percentage of TNBC in the cN0 group (60%) compared with that of the cN1–3 group (40%). This imbalance

is explained by the more preferential treatment of TNBC with NAC than the other subtypes of tumors in case of cN0.

We found that a majority ($n = 8$) of 12 patients with gHRRm had HRD-high tumors consistent with previous reports [4,7,8,10,11,21,32–36]. These eight patients had BRCA1 or BRCA2 germline mutations. The wild-type allele was lost in all tumors. Of the four HRD-low tumors, interestingly, one case with a BRCA2 germline mutation retained the wild-type allele (Suppl. Table 2, Patient 10), whereas the other one lost the mutant allele somatically (Suppl. Table 2, Patient 9). Jonsson et al. reported that 8% of BRCA1 or BRCA2 germline mutation carriers lost the mutant allele somatically in tumors in a pan-cancer cohort [6]. Thus, it is speculated that the above-mentioned two tumors with a BRCA2 germline mutation have low HRD scores because of the retention of a wild-type allele.

Then, we studied the impact of sHRRm on the HRD scores. Although sHRRm was observed in 27 tumors, the corresponding HRD scores were significantly lower than those of gHRRm tumors and similar to those of the HRRw tumors. A difference in somatic mutations between the HRD-core and HRD-related genes showed no significant influence on the HRD scores (Suppl. Fig. 2). These results suggested that sHRRm did not impact on the HRD scores. Contrastingly, Mutter et al. reported that tumors harboring sHRRm with LOH in one of the 93 HRR genes (BRCA1/2 excluded) had significantly higher LST scores than those with the HRRw ($P < 0.001$), indicating the importance of bi-allelic HRR gene inactivation in HRD induction [37]. We then analyzed the impact of the combination of sHRRm and LOH on HRD scores. A significant difference was not observed, however, the HRD scores in the sHRRm tumors with LOH tended to be higher than those without (Suppl. Fig. 2). Recurrent breast cancers with somatic BRCA1 or BRCA2 mutations have been shown to favorably respond to olaparib, suggesting that both germline and somatic mutations induce HRD, increasing the sensitivity to PARP inhibitors [38]. Of note, the LOH status was not been reported in that study. As for somatic BRCA1 or BRCA2 mutations, we only detected a single tumor (somatic BRCA1 mutation), preventing further analysis.

Association of HRD scores with a response to NAC consisting of anthracycline and taxane has mostly been investigated in TNBC, and rarely in the luminal subtype. Thus, we focused on the luminal subtype in

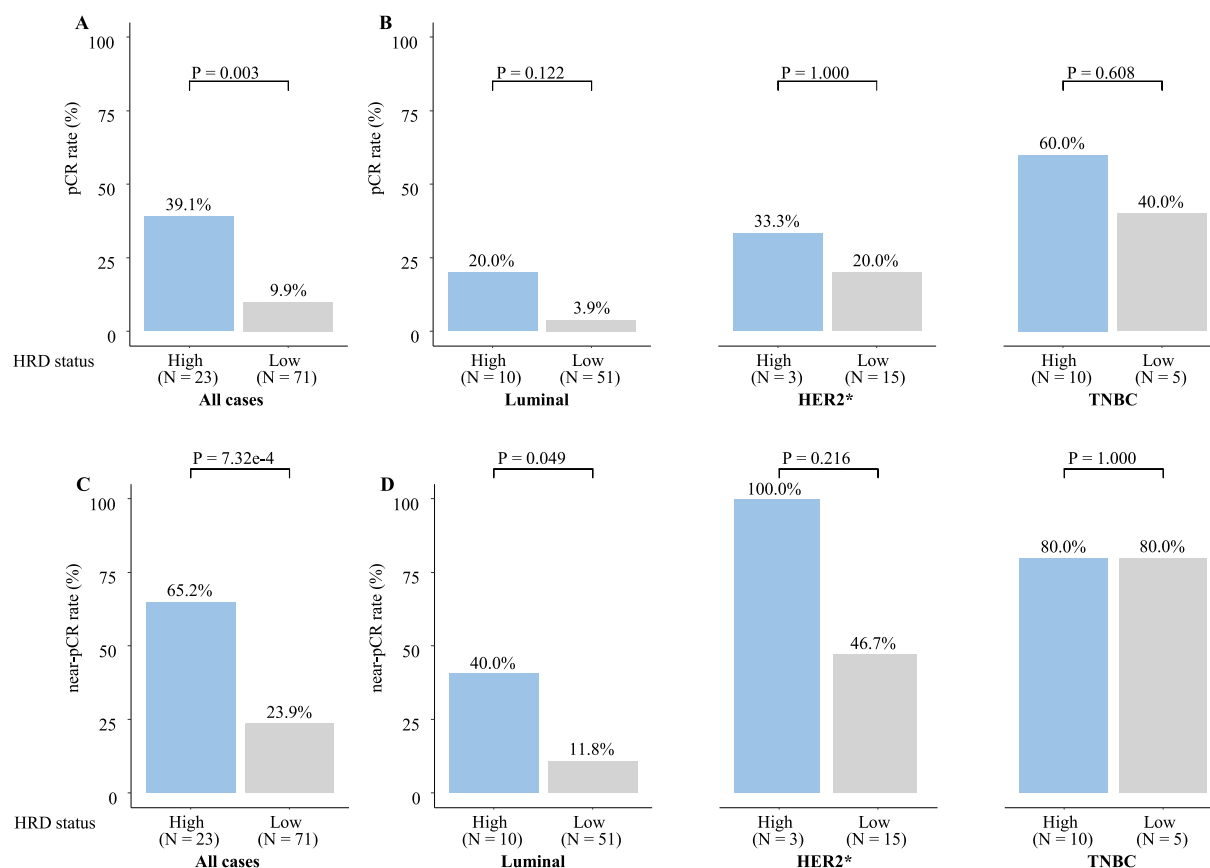


Fig. 4. pCR and near-pCR rates to NAC by HRD status. Patients ($n = 94$) treated with neoadjuvant chemotherapy (NAC) without trastuzumab were analyzed. pCR rates were analyzed in all patients (A) and each subset (B), and near-pCR rates were also analyzed in all patients (C) and each subset (D). HER2* included luminal-HER2 and HER2 subsets in Fig. 2.

the present study. HRD-high tumors were significantly more likely to achieve pCR than HRD-low tumors when all tumors were considered but not in each subset. The reason why HRD scores were not significantly associated with pCR in the luminal subtype may be because the pCR rate was very low (6.6%, 4/61 patients) in this subtype. It seems to be difficult to evaluate the predictive power of HRD scores for their chemosensitivity when employing pCR as the end-point. Therefore, we used the near-pCR instead (near-pCR rates 16.4%, 10/61 patients) as it was thought to represent the chemosensitivity of the luminal subtype more precisely. We showed that the HRD-high tumors were significantly more likely to achieve near-pCR in only the luminal subtype. Univariate analysis of the parameters showed that only HRD scores were significantly associated with near-pCR (Table 3), suggesting that HRD-high luminal tumors were more likely to benefit from P-FEC and that HRD scores might be a clinically useful marker for chemosensitivity based on the luminal subtype. Then, we also analyzed the prognostic value of HRD scores while we found no significant association between HRD scores and relative-free survival or overall survival in the luminal subtype as well as other subtypes (Suppl. Figs. 3 and 4). On the other hand, Loibl et al. and Sharma et al., have demonstrated that HRD-high tumors were significantly associated with not only pCR but also better prognosis compared with HRD-low tumors in TNBC [9, 13]. These findings suggest HRD scores might be a predictive factor but not a prognostic factor in the luminal subtype.

HRDetect scores (composite mutational-signature) have been recently developed to identify BRCA1/2-deficient tumors with increased accuracy compared with HRD scores. They also predict the prognosis and response to platinum-based chemotherapy in advanced breast cancers [39–41]. As HRDetect scores are determined by whole genome se-

quencing (WGS), we were unable to compare HRDetect scores and HRD scores in the present study. Comparison of these two scores for their clinical utility, especially prediction of response to PARP inhibitors and DNA damaging chemotherapeutic agents, is very important and warrants further study.

In conclusion, HRD scores could be determined by WES as accurately as OncoScan®. HRD-high tumors had biologically aggressive phenotypes and were most prevalent in TNBC. gHRRm and not sHRRm was associated with high HRD scores. In the luminal subset, HRD-high tumors were significantly associated with a higher response to neoadjuvant paclitaxel - FEC, suggesting a clinical utility of these scores as a predictor of response to such a chemotherapy. Our results require confirmatory testing in future studies with larger patients cohorts.

Declaration of Competing Interest

Kim S.J. received the honoraria from AstraZeneca, Novartis, Taiho, Eli Lilly, Daiichi Sankyo, Nippon Kayaku, Kyowa Kirin, and Chugai. Naoi Y. received the research funding from Sysmex and AstraZeneca. Kagara N. received the research funding from Novartis. Shimoda M. received the honoraria from Eisai, Takeda, Novartis, and Chugai. Noguchi S. has an advisory role with AstraZeneca, Sysmex, and Nittobo, received the honoraria from AstraZeneca, Novartis, Pfizer, Eli Lilly, and Chugai, and the research funding from Sysmex, Nittobo, Taiho, Chugai, Novartis, Pfizer, and Daiichi Sankyo, and holds the joint patents unrelated to this study with Sysmex. Shimazu K. received the honoraria from AstraZeneca, Eisai, Novartis, Pfizer, Eli Lilly, Daiichi Sankyo, Sysmex and Chugai. Sota Y., Tanei T., and Miyake T. have no conflict of interest.

CRediT authorship contribution statement

Seung Jin Kim: Conceptualization, Investigation, Data curation, Writing - original draft, Writing - review & editing. **Yoshiaki Sota:** Data curation, Writing - original draft. **Keiichiro Honma:** Investigation.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.tranon.2020.100986.

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Table S1 HRR genes analyzed by whole exome sequencing

Gene	Alias	Category	Process
BABAM1	NBA1, Merit40	core	HR
BARD1		core	HR
BLM		core	HR
BRCA1		core	HR
BRCA2	FANCD1	core	HR, FA
BRCC3	BRCC36	core	HR, Deubiquitylation
BRIP1	FANCI, BACH1	core	HR, FA
EME1	MMS4L	core	HR
EME2		core	HR
EXO1		core	HR
ABRAXAS1	Abraxas, ABRA1, FAM175A	core	HR
GEN1		core	HR
H2AFX	H2AX	core	DSB Signaling
MDC1		core	DSB Signaling
MRE11	MRE11A	core	DSB Signaling, HR
MUS81		core	HR
NBN	NBS	core	DSB Signaling, HR
PALB2	FANCN	core	HR, FA
RAD50		core	DSB Signaling, HR
RAD51		core	HR
RAD51AP1		core	HR
RAD51B	RAD51L1	core	HR
RAD51C	FANCO, RAD51L2	core	HR, FA
RAD51D	RAD5L3	core	HR
RAD52		core	HR
RAD54B		core	HR
RAD54L		core	HR
RBBP8	CTIP	core	HR
REV1	REV1L	core	HR
RPA1		core	HR, NER
RPA2		core	HR, NER
RTEL1		core	HR
SEM1	DSS1; SHFM1	core	HR
SLX1A	SLX1; GIYD1	core	HR
SLX1B	GIYD2	core	HR
SLX4	FANCP	core	HR, FA
TP53BP1	53BP1	core	DSB Signaling
UIMC1	RAP80	core	HR
XRCC2		core	HR
XRCC3		core	HR
ABL1		related	DDR
ATM		related	DSB Signaling
ATR		related	DSB Signaling
ATRIP		related	DDR

BAP1		related	DDR
BRE	BRCC4	related	DDR
FAAP100	C17orf70	related	FA
FAAP24	C19orf40	related	HR, FA
FAAP20	C1orf86	related	FA
CDK12		related	DDR
CHEK1		related	DSB Signaling
CHEK2		related	DSB Signaling
DMC1		related	HR
DNA2		related	HR
DNMT3A		related	
ERCC4	XPF, FANCC	related	FA, SSA, NER
FAN1	MTMR15	related	FA
FANCA		related	FA
FANCB		related	FA
FANCC		related	FA
FANCD2		related	FA
FANCE		related	FA
FANCF		related	FA
FANCG	XRCC9	related	FA
FANCI	KIAA1794	related	FA
FANCL		related	FA
FANCM		related	FA
FEN1		related	DDR
FIGNL1		related	HR
HELQ		related	HR
KAT5	tip60	related	chromatin: HAT
NONO	NRB54; p54	related	HR
PARP1	ADPRT	related	SSBR
PIAS1		related	chromatin: sumoylation
PIAS4		related	chromatin: sumoylation
POLN	POLP4	related	Polymerase
POLQ		related	Polymerase
RAD1		related	DDR
RAD17	RAD24	related	DDR
RAD9A		related	DDR
RECQL	RECQ1	related	DDR
RECQL4		related	DDR
RIF1		related	HR
RMI1		related	HR
RMI2		related	HR
RNF168		related	DSB Signaling
RNF169		related	DSB Signaling
RNF4		related	Ubiquitination
RPA3		related	NER
SETMAR	Metnase	related	
SFPQ	PSF	related	

SFR1		related	HR
SMC1A	SMC1	related	cohesin
SMC3		related	cohesin
SMC5		related	cohesin
SPIDR	KIAA0146	related	HR
TIMELESS		related	
TIPIN		related	DDR
TOPBP1		related	DDR
USP1		related	FA
USP11		related	HR, ubiquitination
WRN		related	DDR

FA: Fanconi Anemia; DSB Signaling: Double-strand break signaling; DDR: DNA Damage Response;
NER: nucleotide excision repair; SSBR: single-strand break repair

Table S2 Germline HRR gene mutations

ID	Chr	Start	End	Ref	Alt	Gene	Variant type	CLNSIG	Serum		Tumor		HRD	LOH
									AD	AF	AD	AF		
Patient1	chr17	43099817	43099817	G	A	BRCA1	stopgain	Pathogenic	41,51	0.554	31,66	0.680	83	+
Patient2	chr17	43063369	43063370	CA	-	BRCA1	frameshift deletion	Pathogenic	14,9	0.391	13,11	0.458	85	+
Patient3	chr13	32379885	32379885	-	A	BRCA2	frameshift insertion	Pathogenic	10,34	0.773	19,67	0.779	75	+
Patient4	chr13	32339764	32339765	TG	-	BRCA2	frameshift deletion	Pathogenic	17,26	0.605	18,38	0.679	49	+
Patient5	chr13	32331004	32331008	CAAAT	-	BRCA2	frameshift deletion	Pathogenic	8,4	0.333	5,6	0.545	62	+
Patient5	chr8	94399540	94399540	C	A	RAD54B	nonsynonymous SNV	Pathogenic	26,24	0.480	71,69	0.493	62	-
Patient6	chr13	32339929	32339932	AAT	-	BRCA2	frameshift deletion	Pathogenic	21,26	0.553	17,56	0.767	67	+
Patient7	chr13	32346841	32346841	C	T	BRCA2	stopgain	Pathogenic	12,3	0.200	12,19	0.613	47	+
Patient8	chr13	32346841	32346841	C	T	BRCA2	stopgain	Pathogenic	15,18	0.545	7,9	0.563	51	+
Patient9	chr13	32346841	32346841	C	T	BRCA2	stopgain	Pathogenic	14,17	0.548	23,4	0.148	29	+
Patient100	chr13	32333284	32333284	A	-	BRCA2	frameshift deletion	Pathogenic	7,10	0.588	20,20	0.500	28	-
Patient11	chr16	23621408	23621408	G	A	PALB2	stopgain	Pathogenic	17,26	0.605	47,51	0.520	39	+
Patient12	chr8	94399540	94399540	C	A	RAD54B	nonsynonymous SNV	Pathogenic	38,21	0.356	32,45	0.584	26	+

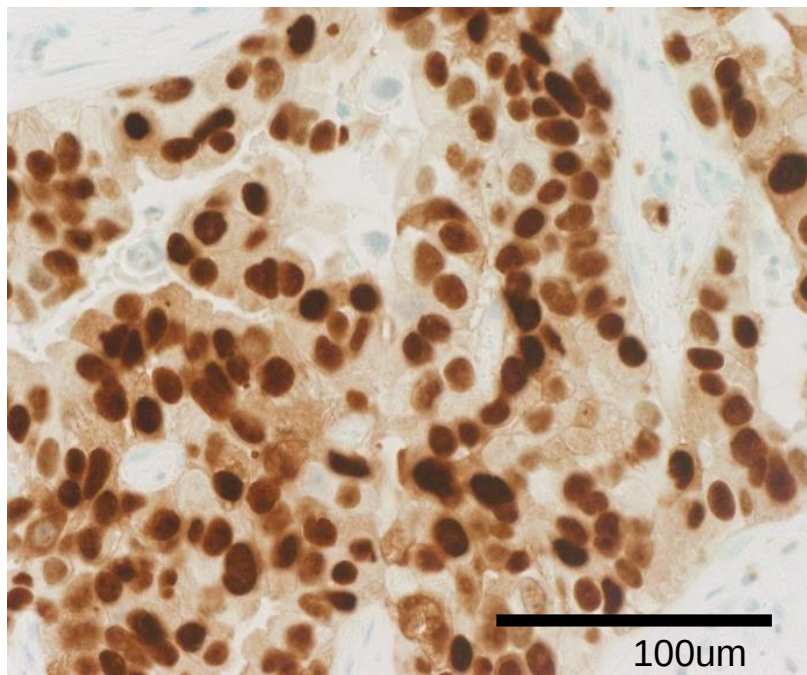
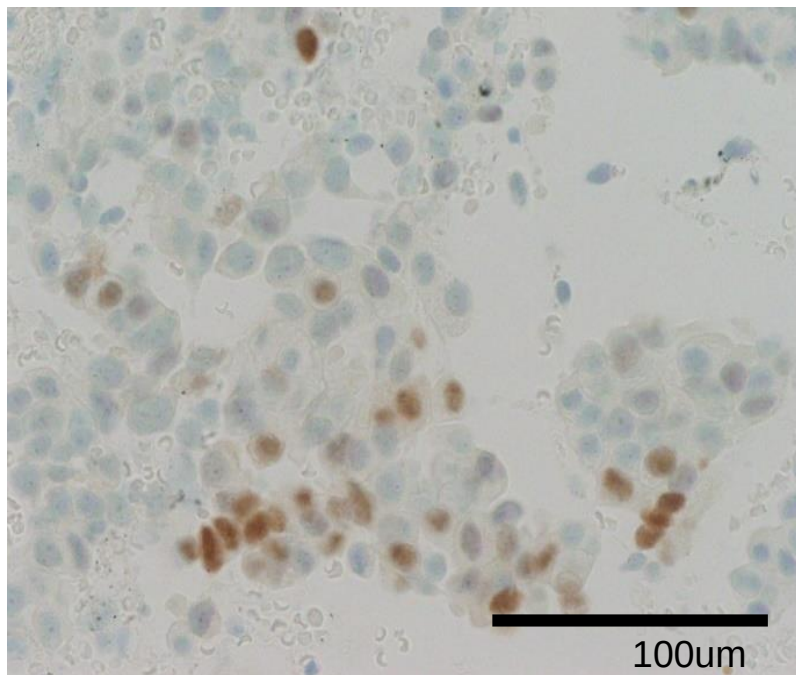
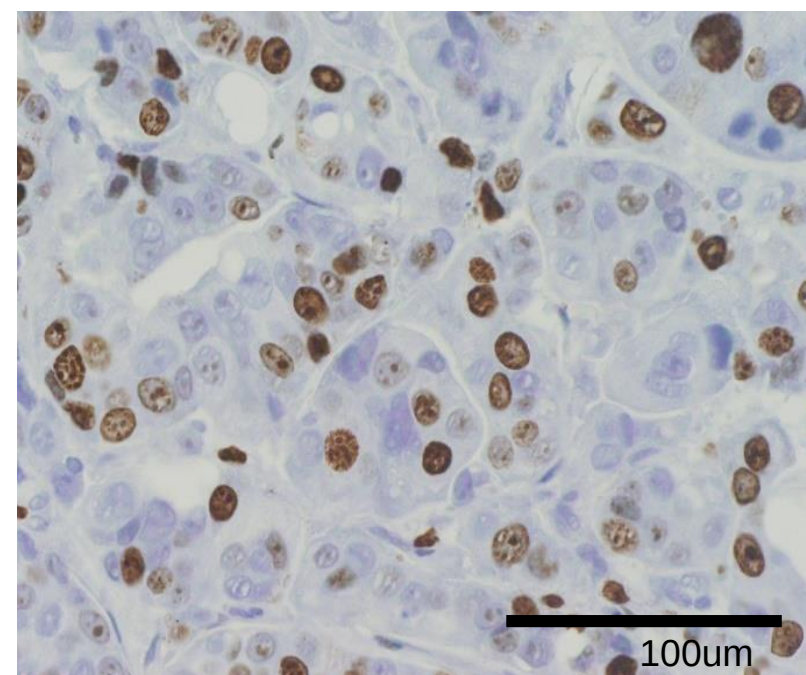
CLNSIG, clinical significance; AD, allele depth; AF, allele frequency; DP, Depth; GQ, Quality of the assigned genotype

Table S3: Somatic HRR gene mutations

ID	Chr	Start	End	Ref	Alt	Gene	Variant type	CLNSIG	AD	AF	HRD	LOH	
Patient13	chr11	108321293	108321302	GCTACTAGAG	-	ATM	frameshift deletion		59,9	0.130	3	-	
Patient14	chr11	108316019	108316019		C	T	ATM	nonsynonymous SNV	81,6	0.078	12	-	
Patient15	chr11	108251029	108251029		G	C	ATM	nonsynonymous SNV	41,22	0.354	39	+	
Patient16	chr3	142469454	142469454		C	G	ATR	nonsynonymous SNV	23,8	0.273	59	+	
Patient16	chr17	61683819	61683819		G	A	BRIP1	nonsynonymous SNV	US	77,11	0.133	59	+
Patient16	chr17	81551048	81551048		C	T	FAAP100	nonsynonymous SNV	407,156	0.278	59	+	
Patient16	chrX	14850589	14850589		G	C	FANCB	stopgain	49,16	0.254	59	-	
Patient16	chr8	144515821	144515821		C	G	RECQL4	nonsynonymous SNV	160,63	0.284	59	-	
Patient16	chr3	133608923	133608923		G	A	TOPBP1	stopgain	45,14	0.246	59	-	
Patient17	chr3	142493248	142493248		C	A	ATR	stopgain	57,6	0.107	50	-	
Patient18	chr3	48447042	48447042		T	A	ATRIP	nonsynonymous SNV	67,8	0.117	77	+	
Patient19	chr17	43093458	43093458		T	A	BRCA1	nonsynonymous SNV	77,22	0.228	22	+	
Patient20	chr17	39462456	39462456		G	A	CDK12	nonsynonymous SNV	75,7	0.089	5	+	
Patient20	chr9	35079468	35079468		C	G	FANCG	nonsynonymous SNV	215,21	0.092	5	-	
Patient20	chr3	4313693	4313693		G	C	SETMAR	nonsynonymous SNV	135,13	0.093	5	-	
Patient21	chr22	28734714	28734714		C	A	CHEK2	nonsynonymous SNV	US	89,6	0.072	30	-
Patient21	chr1	35192490	35192490		G	T	SFPQ	nonsynonymous SNV	46,5	0.101	30	+	
Patient22	chr16	13948099	13948116	TCTGAAACCC TTCCCGAG	-	ERCC4	nonframeshift deletion		143,11	0.077	15	-	
Patient23	chr1	241866872	241866872		C	T	EXO1	stopgain	98,7	0.074	17	-	
Patient24	chr9	35075493	35075502	GGGCACCCAG	-	FANCG	frameshift deletion		285,17	0.059	16	-	
Patient25	chr14	45167019	45167019		C	A	FANCM	nonsynonymous SNV	33,6	0.170	36	-	
Patient25	chr8	144512230	144512230		C	G	RECQL4	nonsynonymous SNV	302,31	0.095	36	-	
Patient25	chr2	151463251	151463251		C	T	RIF1	nonsynonymous SNV	59,8	0.130	36	-	
Patient25	chrX	47247887	47247887		C	T	USP11	nonsynonymous SNV	58,27	0.322	36	-	
Patient26	chr11	65860989	65860989		G	C	MUS81	nonsynonymous SNV	187,12	0.065	2	-	
Patient27	chr11	65860885	65860885		G	C	MUS81	nonsynonymous SNV	157,18	0.107	44	+	
Patient27	chr5	132640744	132640744		G	C	RAD50	nonsynonymous SNV	90,9	0.099	44	+	
Patient28	chrX	71291848	71291848		G	A	NONO	nonsynonymous SNV	61,8	0.127	37	-	
Patient29	chr16	23636244	23636244		T	C	PALB2	nonsynonymous SNV	106,20	0.164	32	-	

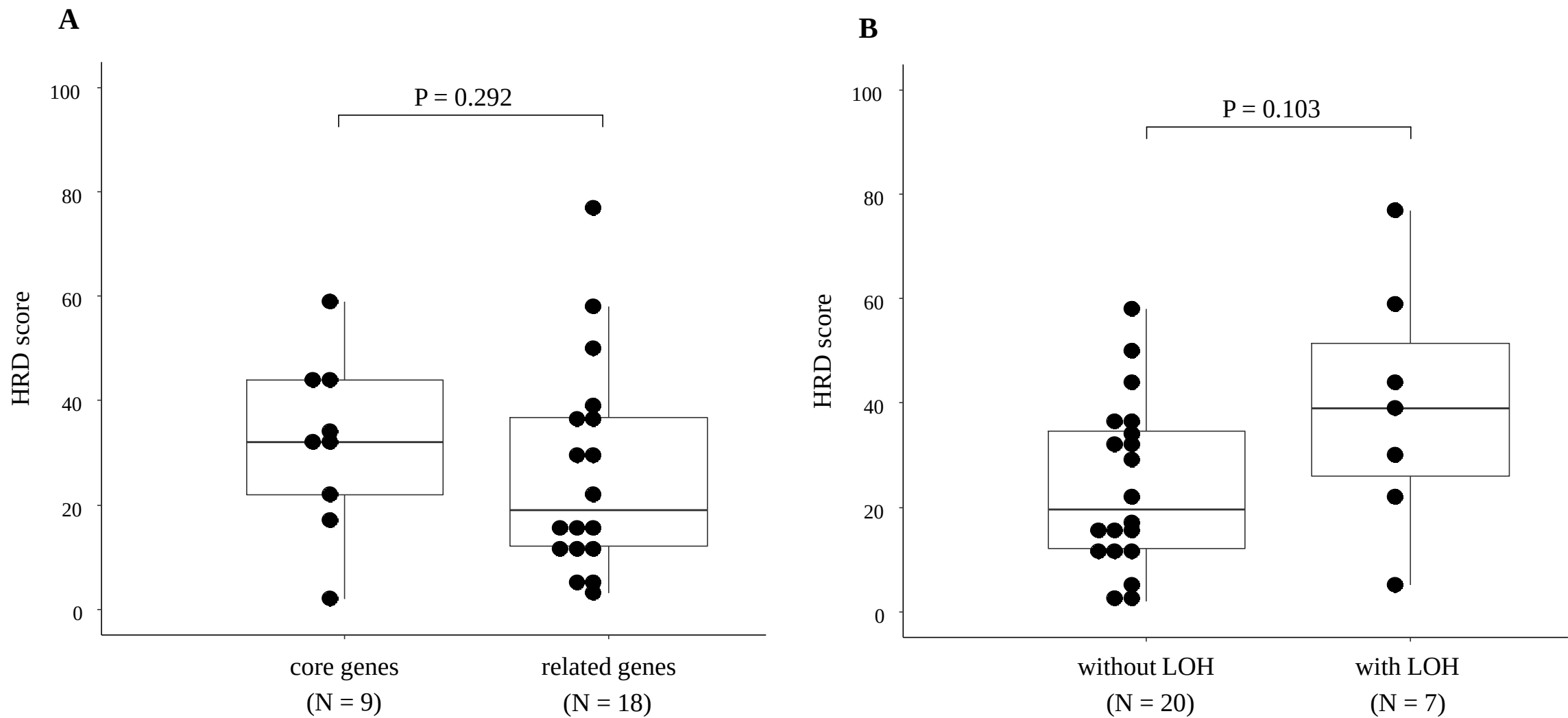
Patient30	chr3	121440114	121440114	T	C	POLQ	nonsynonymous SNV		26,6	0.205	11	-
Patient31	chr3	121487393	121487402	GCACCGCCAC	-	POLQ	frameshift deletion		87,7	0.083	29	-
Patient32	chr14	67865080	67865080	G	A	RAD51B	nonsynonymous SNV		54,9	0.153	32	-
Patient33	chr8	144511946	144511946	C	T	RECQL4	nonsynonymous SNV	US	191,51	0.209	22	-
Patient33	chrX	53405309	53405309	C	T	SMC1A	nonsynonymous SNV		175,59	0.254	22	-
Patient34	chr8	144517455	144517455	C	T	RECQL4	nonsynonymous SNV		153,13	0.083	58	-
Patient35	chr8	144514071	144514071	C	A	RECQL4	nonsynonymous SNV		216,36	0.146	15	-
Patient36	chr16	3597489	3597489	G	A	SLX4	unknown		86,17	0.171	44	-
Patient37	chr3	133656762	133656770	GCTACCAAC	-	TOPBP1	nonframeshift deletion		113,6	0.053	5	-
Patient38	chr15	43477599	43477599	T	C	TP53BP1	nonsynonymous SNV		61,15	0.206	34	-
Patient39	chr8	31067090	31067090	C	G	WRN	nonsynonymous SNV		59,5	0.090	12	-

CLNSIG, clinical significance; AD, allele depth; AF, allele frequency; DP, Depth; AMBQ, median base quality of alternate alleles; US, Uncertain significance

A**ER****B****PR****C****Ki-67**

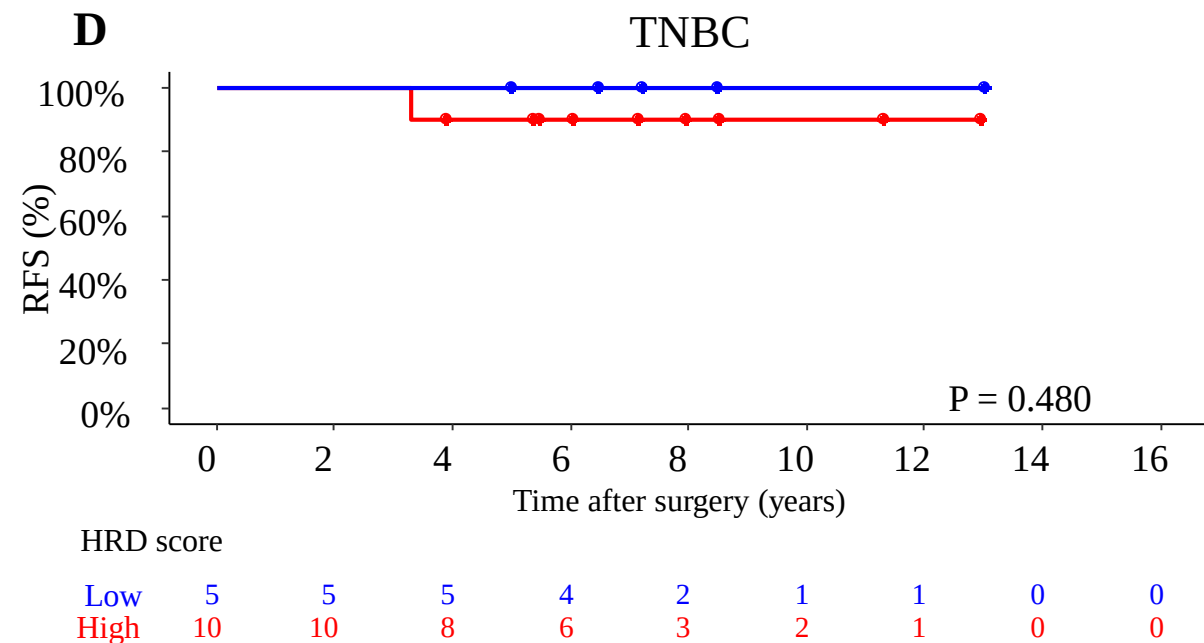
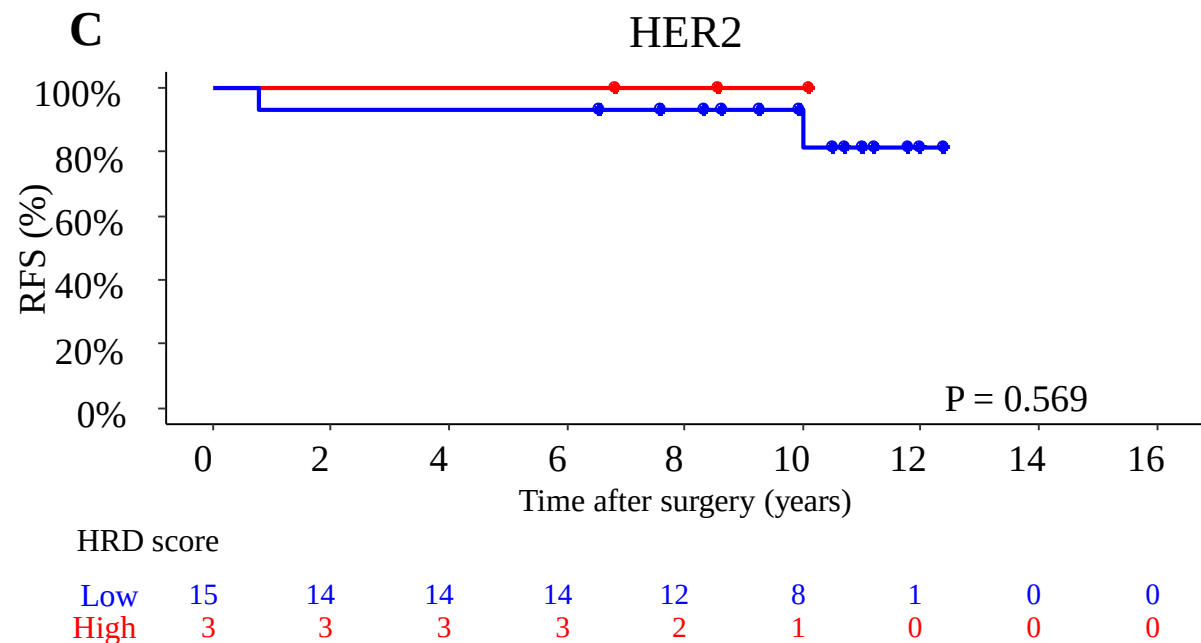
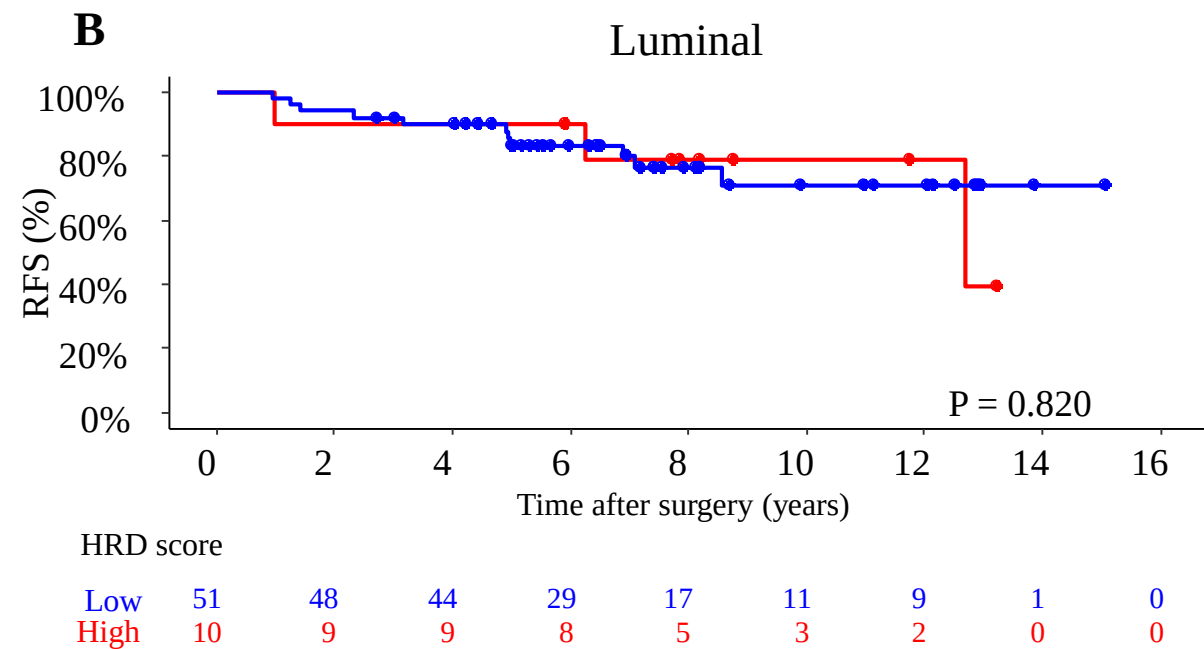
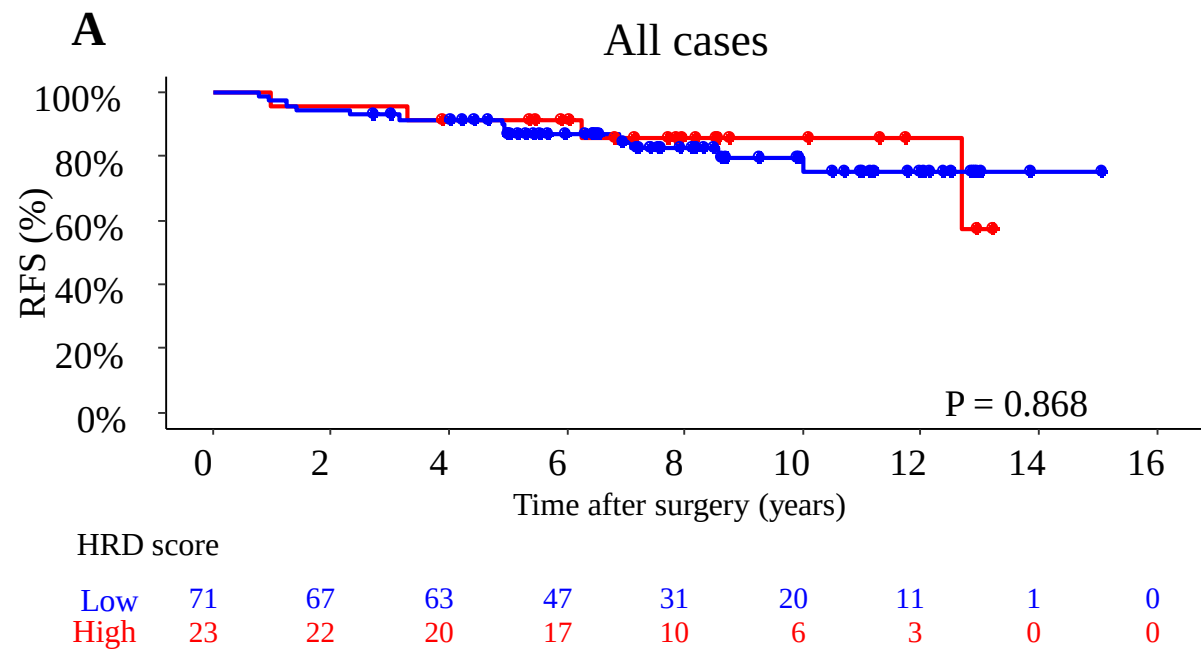
Supplementary Figure 1. Representative immunohistochemical staining of ER, PR, and Ki-67 in breast tumors.

ER was strongly stained ($> 66.7\%$) (A). PR was weakly stained ($10\% \leq < 33.3\%$) (B). Ki-67 was moderately stained (40-50%) (C).



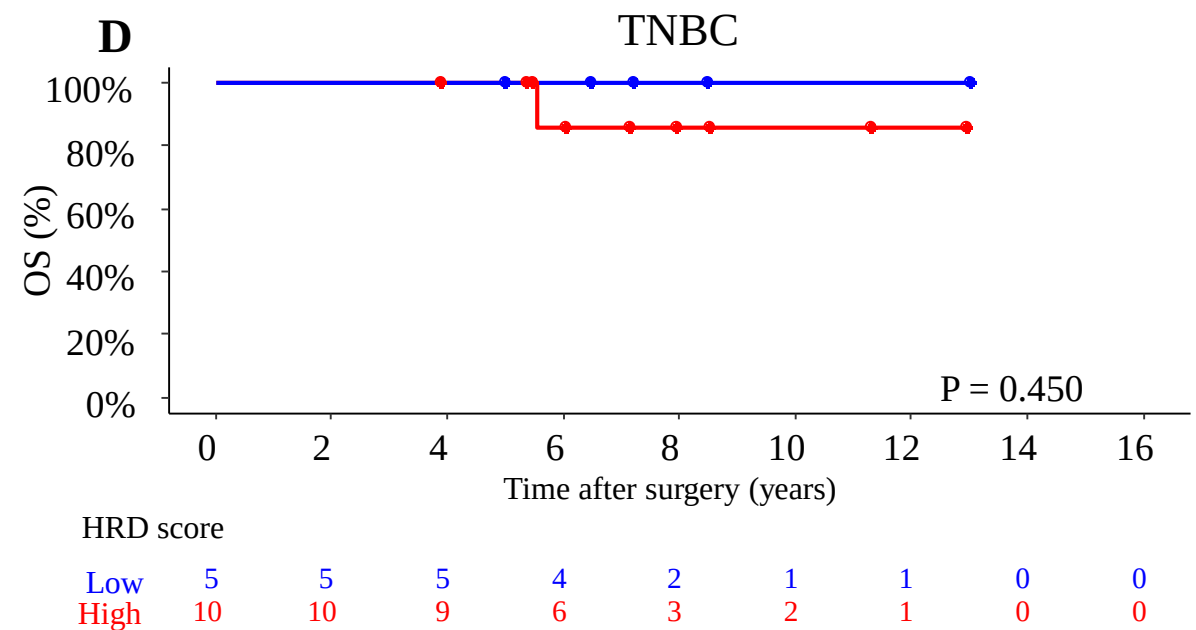
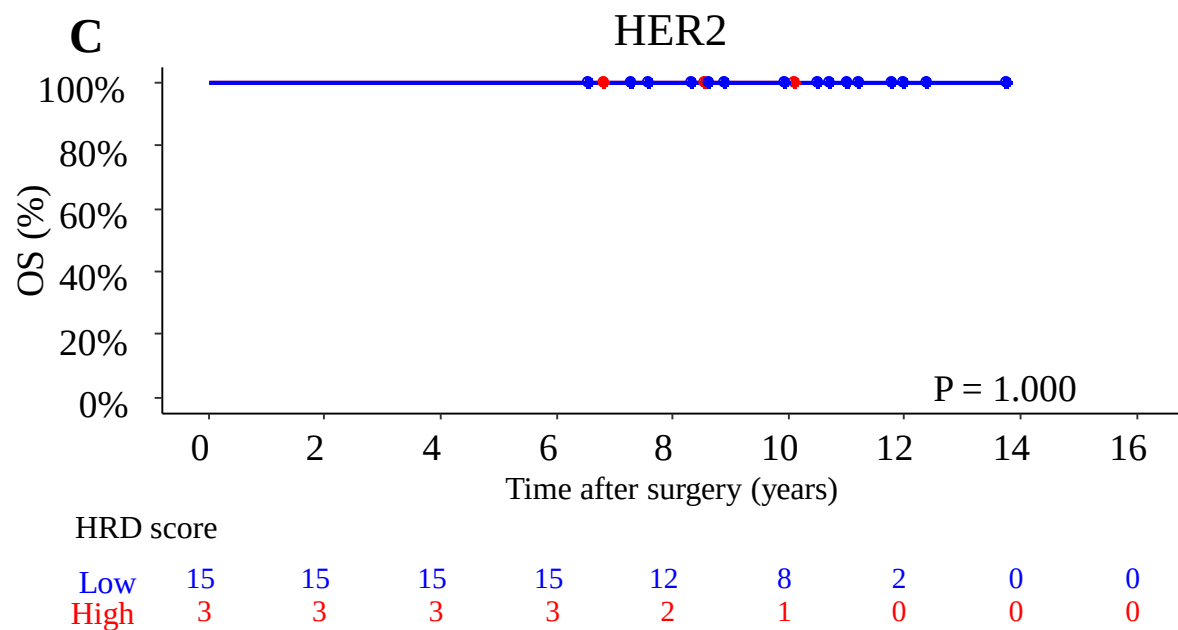
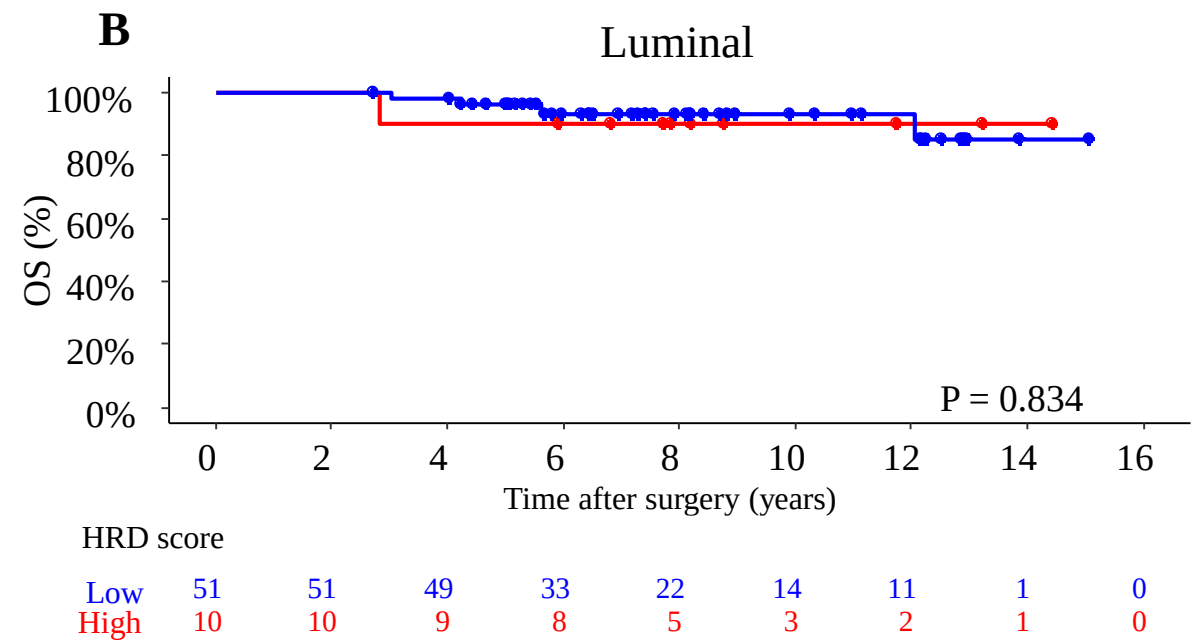
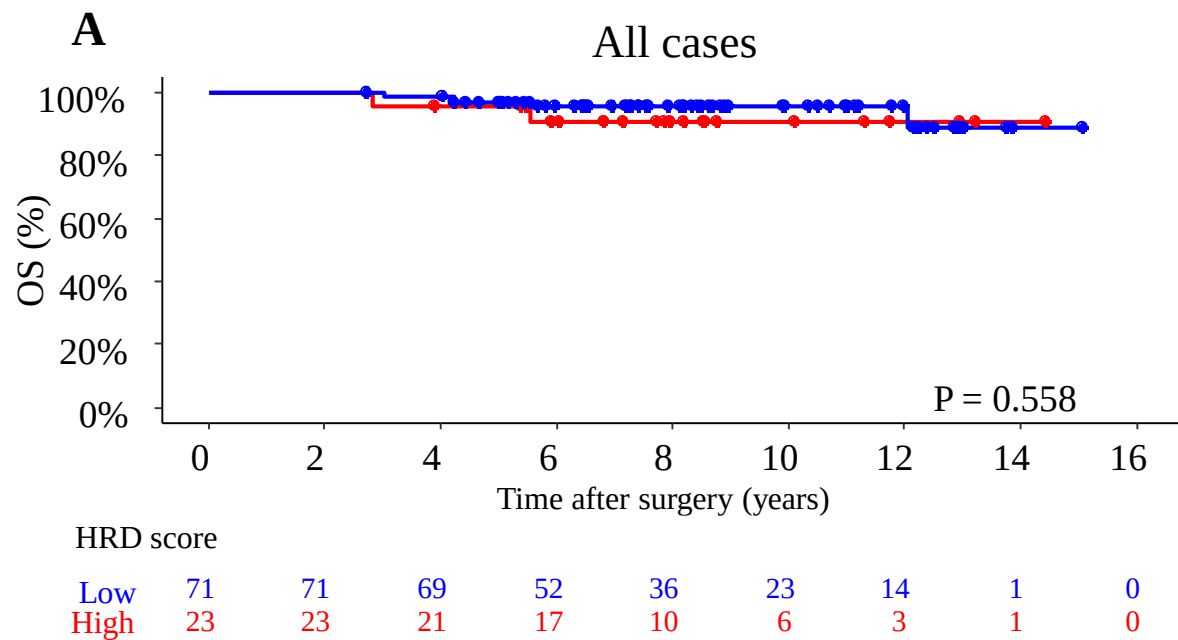
Supplementary Figure 2. HRD scores according to somatic HRR gene mutations

Somatic HRR genes were divided into the HRR-core and HRR-related genes according to their function. The HRD scores were compared between tumors with somatic mutations in HRR-core and HRR-related genes (A). In tumors with somatic HRR gene mutations, LOH at the mutated gene loci were also analyzed. The HRD scores were compared between somatic HRR gene mutant tumors with and without LOH (B).



Supplementary Figure 3. Relapse-free survival curves according to HRD status

Patients (n = 94) treated with neoadjuvant chemotherapy (NAC) without trastuzumab were analyzed. Relapse-free survival (RFS) was analyzed using the Kaplan-Meier method in all patients (A), the luminal subtype (B), the HER2 type included luminal-HER2 and HER2 subsets in Figure 2 (C), and TNBC (D). The differences were assessed with the log-rank test. Relapse didn't include a contralateral breast in this analysis. The median follow-up period was 6.86 (0.34 - 15.06) months.



Supplementary Figure 4. Overall survival curves according to HRD status

Patients (n = 94) treated with neoadjuvant chemotherapy (NAC) without trastuzumab were analyzed. Overall survival (OS) was analyzed using the Kaplan-Meier method in all patients (A), the luminal subtype (B), the HER2 type included luminal-HER2 and HER2 subsets in Figure 2 (C), and TNBC (D). The differences were assessed with the log-rank test. The median follow-up period was 7.25 (2.26 - 15.06) months. During this period, there were six events (four breast cancer-caused deaths and two other-caused deaths).